

S475/1
SUBSIDIARY
MATHEMATICS

Paper 1
Nov./Dec. 2025
 $2\frac{2}{3}$ hours



UGANDA NATIONAL EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

SUBSIDIARY MATHEMATICS

Paper 1

2 hours 40 minutes

INSTRUCTIONS TO CANDIDATES:

*This paper consists of **two** Sections; **A** and **B**.*

*Section **A** is **compulsory**.*

*Answer only **four** questions from Section **B**, choosing **at least one** question from each part.*

*Any additional question(s) answered will **not** be marked.*

*All necessary working **must** be shown clearly.*

Begin each answer on a fresh sheet of paper.

Graph paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formulae may be used.

SECTION A (40 MARKS)*This section is Compulsory.*

1. Find $\int \sin^2 \theta \cos \theta \, d\theta$. (05 marks)

2. Use the matrix method to solve the following simultaneous equations:

$$3x + 4y = 2$$

$$5x - 6y = 16$$

(05 marks)

3. Solve the equation:

$$\cos 2x - 4 \sin 2x = 0 \text{ for } 0^\circ \leq x \leq 180^\circ.$$

(05 marks)

4. A team of 5 players is to be selected from 7 boys and 6 girls. If the team must have more girls than boys, in how many ways can the team be selected?

(05 marks)

5. The table below shows the price in thousands of Uganda Shillings (Ugsh) and quantities of four items A , B , C and D for the years 2019 and 2020.

Item	Price per unit ('000Ugsh)		Quantities	
	2019	2020	2019	2020
A	100	120	36	42
B	110	100	96	88
C	50	65	10	12
D	80	85	11	10

(a) Using 2019 as the base year, calculate the value index of the items.

(04 marks)

(b) Comment on your result obtained in (a).

(01 mark)

6. Events A and B are such that $P(B) = 0.2$, $P(A \cap B) = 0.08$ and $P(B/A) = 0.4$. Find the value of:

(a) $P(A)$

(03 marks)

(b) $P(A \cup B)$

(02 marks)

7. A discrete random variable X has the following probability distribution.

x	1	2	3	4
$P(X=x)$	b	0.35	0.40	0.05

Determine the

(a) value of b .

(02 marks)

(b) $P(1 \leq X < 3)$

(03 marks)

12. The rate of increase in the quantity of a certain commodity demanded is inversely proportional to the quantity, N , of the commodity present at any time, t , minutes. Initially the quantity of the commodity was 300 and after 5 minutes it was 350.
- (a) (i) Form a differential equation for the rate of increase in the quantity of the commodity demanded. (02 marks)
- (ii) Solve the differential equation. (10 marks)
- (b) Calculate the quantity of the commodity present after 20 minutes. (03 marks)

Part Two: Statistics

13. The table below shows marks scored by 70 students in a test.

Marks	Number of students
20 – 29	05
30 – 39	11
40 – 49	15
50 – 59	22
60 – 69	09
70 – 79	04
80 – 89	04

- (a) Calculate the:
- (i) mean mark. (04 marks)
- (ii) standard deviation. (03 marks)
- (iii) Median mark. (03 marks)
- (b) (i) Draw a histogram for the given data.
- (ii) Use your histogram in (b)(i) to estimate the modal mark. (05 marks)
14. The quarterly sales of animals in a market from 2021 to 2023 are given in the table below.

Year	Quarter			
	1 st	2 nd	3 rd	4 th
2021	55	67	50	72
2022	62	80	73	91
2023	65	94	72	95

- (a) Calculate the four-point moving averages for the data. (04 marks)

- (b) (i) On the same axes plot the graphs of the given original data and the four-point moving averages.
(ii) Comment on the trend of sales of animals over the given period of time. (07 marks)
- (c) Use your graph to estimate the number of animals sold in the 1st quarter of 2024. (04 marks)

15. The intelligence quotient (IQ) of 12 mothers (x) and that of their respective daughters (y) were investigated and recorded as shown in the table below.

Mother's IQ (x)	160	153	154	154	149	145	148	144	140	138	135	134
Daughter's IQ (y)	159	150	156	149	156	153	154	141	140	148	144	135

- (a) (i) Plot a scatter diagram for the given data. (03 marks)
(ii) Draw a line of best fit on the scatter diagram. (02 marks)
(iii) Use the diagram to estimate the IQ of a mother whose daughter's IQ is 170. (02 marks)
- (b) (i) Calculate the rank correlation coefficient for the data. (06 marks)
(ii) Comment on your result. (02 marks)

16. A continuous random variable X has a probability density function (pdf) given by:

$$f(x) = \begin{cases} \frac{1}{k}x^2; & 0 \leq x < 3 \\ \frac{9}{k}; & 3 \leq x \leq 5 \\ 0; & \text{elsewhere;} \end{cases}$$

Determine the;

- (a) Value of k . (05 marks)
(b) Expectation of X , ($E(X)$). (05 marks)
(c) $P(2 \leq x \leq 4)$. (05 marks)

8. A survey carried out on a certain farm, revealed that 15 % of the animals vaccinated against a certain disease actually developed the disease. Given that a sample of 10 animals is drawn at random from the farm, determine the probability that between 3 and 7 animals have the disease. (05 marks)

SECTION B (60 MARKS)

Answer only **four** questions from this section, with **at least one** question from each part. All questions carry equal marks.

Part One: Pure Mathematics

9. (a) A particle moves in a straight line such that its distance (x m) from the origin at t seconds is given by $x = 45t + 11t^2 - t^3$.
Find the;
(i) velocity of the particle after 9 seconds. (03 marks)
(ii) acceleration of the particle after 3 seconds. (03 marks)
- (b) The equation of a curve is given by $y = 5x^2 - 30x + 24$.
Determine the;
(i) coordinates of the stationary point on the curve. (06 marks)
(ii) nature of the stationary point on the curve. (03 marks)
10. (a) Without using a calculator, show that:
$$\sqrt{\frac{1}{8}} + \sqrt{\frac{1}{12}} - \sqrt{\frac{1}{27}} = \frac{1}{4}\sqrt{2} + \frac{1}{18}\sqrt{3}$$
 (07 marks)
- (b) Solve the equation $2^{2x} - 7(2^x) + 12 = 0$ (08 marks)
11. A wholesaler wishes to transport 870 bags of posho from a milling factory using a lorry and a pickup. The lorry can carry 150 bags per trip while the pickup can carry 60 bags per trip. The cost of each trip for the lorry is Shs25,000 while for the pickup is Shs22,000. The pickup makes more trips than the lorry. The amount of money available for transporting posho is Shs220,000.
- (a) If x represents the number of trips made by the pickup and y represents the trips made by the lorry, write **three** inequalities representing the given information. (03 marks)
- (b) On the same axes, plot graphs for the inequalities by shading the unwanted regions. (07 marks)
- (c) Determine the number of trips that the lorry and the pickup should make in order to minimise the transport cost. (05 marks)